Qui Loc PHAN

Konatus | phanquiloc001@gmail.com | locphan2001.github.io | Vietnamese

Education

Université Paris Dauphine & École des Mines de Paris **2023 – 2025**

Master of Computer Science *Paris, France*

- Graduated with highest grade in Combinatorial Optimization specialization of program
- Awarded Excellence Scholarship from partner institution ENS Paris Saclay and ENS Paris for 2 consecutive years

Ho Chi Minh University of Education **2019 – 2023**

Honours Bachelor of Mathematics *Ho Chi Minh, Vietnam*

- Graduated with Valedictorian Achievement (rank 1st/150)

Industrial Experience

Konatus, Startup of École Polytechnique **Sep 2025 – present**

Research Engineer *Palaiseau, France*

- Research scheduling and resource allocation optimization problems for large enterprise clients managing complex projects
- Develop mathematical models incorporating complex real-world constraints and multiple sources of uncertainty
- Improve internal planning tools using evolutionary computation and machine learning to enhance decision efficiency
- Strengthen robustness and flexibility of solution by modeling uncertainty, adapting to resource variability and disruptions
- Evaluate and benchmark the company's solutions under different uncertainty models and operating conditions
- Collaborate with academic researchers to explore and integrate state-of-the-art methods in operations research
- Work hand-in-hand with industry experts to deploy and validate methods on real historical and operational datasets

Corporate Research & Sciences Department, Dassault Systèmes **Mar 2025 – Sep 2025**

Research Engineer Intern *Vélizy-Villacoublay, France*

- Study quantum algorithms for constraint programming: optimization problems with integer and logical constraints
- Develop and implement for practical use cases in long-term, explore near-term challenges, conduct comparative analysis
- Practical case study on Healthcare Staff Scheduling at Paris Hospital with multi-objectives and multi-skills

Academic Experience

Optimization and Control Group, UMA, ENSTA Paris **May 2024 – Sep 2024**

Graduate Research Intern *Palaiseau, France*

- Study how to optimize resources used in quantum circuit with two objectives: number of gates and depth of circuit
- Understand and construct mathematical programming model and graph optimization model for the problems
- Design heuristics algorithms that produce good solutions and benchmark the results with available open-source libraries

HCMUE & Vietnam Academy of Science and Technology **10 months**

Undergraduate Research Intern *Ho Chi Minh, Vietnam*

- Study fundamental aspects of combinatorics and abstract algebra, model and address combinatorial problems
- Design or develop, and evaluate algebraic algorithms both in theory and practice mathematically

Project

Production Problem **2024**

- Construct mathematical programming model for the resource allocation problem, inspired from the Flowshop Scheduling
- Implement Campbell - Dudek - Smith algorithm, with variant on preprocessing stage and postprocessing stage
- Perform clustering algorithms on preprocessing and local search methods on postprocessing to improve the solution

Quantum Optimization **2023**

- Encode the mathematical programming model of Max-Cut and Maximum Independent Set into quantum computers
- Code end-to-end QAOA algorithms for the problem, test performance with different number of layers and parameters
- Benchmark with evolutionary algorithms on cost function and introduce some heuristic function for better solutions

Technical Skill

Language: Vietnamese (Native), English (C1), French (B1)

Programming Language: Python (3 years), C++ (2 years), Matlab (2 years)

Optimization Solvers: Gurobi, CPLEX, Google OR-Tools

Quantum Frameworks: Qiskit, PennyLane, Cirq

Updated: October 2025